

Standing Localization under Realized Persistence

A Repair-Localization Theorem under AASC

Amos Jay Maley

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Abstract

This article establishes a fixed-domain standing-localization theorem under AASC. Established AASC closure and induction results exclude the null admissible regime, induce non-vacuous governance from non-degenerate continuation, induce continuation-locus equivalence classes, fix role-anchor equivalence classes, and induce realized persistence-regime equivalence classes from irreversible standing-bearing continuation. The theorem addresses a fixed-domain structural question: once realized persistence is in force, can standing remain global, diffuse, abstract, unbounded, trace-level, or nonlocal while still retaining standing-bearing consequence under lawful repair semantics?

The answer is no. A realized persistence regime retains standing-bearing consequence under irreversible continuation. Lawful repair semantics cannot remain global, external, selector-driven, envelope-drifting, identity-resetting, or unrestrictedly reindexable without violating fixed-domain standing conservation, AMetric non-selection, no same-act repair, or quotient stability. Repair semantics requires bounded repair locality: a bounded role locus over which standing-relevant degradation, correction, identity retention, failure, and continued standing are jointly evaluated. Standing therefore localizes into a bounded standing-localization equivalence class. This class is not a concrete object, physical body, biological organism, observer, material system, or spacetime region. It is the minimal bounded role-structural locus through which standing-bearing consequence retention is preserved under perturbation and lawful repair.

The organizing theorem is

$$\text{RealPers}_D \wedge \text{RepairSem}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

The perturbation-and-repair form follows as

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RepairCap}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

The normal form shows that every admissibility-preserving localization regime factorizes through realized persistence-regime equivalence, standing-bearing consequence retention, perturbation exposure, bounded repair locality, repair semantics, bounded identity locus, and bounded-localization equivalence. The result is a localization theorem rather than a biological, physical, embodied, or concrete-ontology theorem.

Canonical one-line theorem. Realized standing-bearing persistence with lawful repair semantics cannot remain global, diffuse, abstract, or trace-level; repair locality induces a bounded standing-localization equivalence class.

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1 Scope and Target Discipline

This article proves a fixed-domain standing-localization theorem. It does not prove concrete physical reality, spacetime, matter, organisms, observers, biological life, empirical measurement, or material embodiment. Its target is narrower: the bounded standing-localization equivalence class induced when realized persistence is governed by lawful repair semantics.

The word *bounded* is role-structural. It means that standing-bearing consequence retention has a fixed admissibility-relevant locus over which perturbation, correction, identity-memory, failure, and repair are distinguished. It does not mean physical extension, spatial enclosure, biochemical membrane, organismic body, or empirical localization. The phrase *bounded localization structure* is role-structural. It names the minimally bounded locus through which standing-bearing persistence is preserved under perturbation and lawful repair; it is not an ontic bearer.

1.1 Licensed readings

The following readings are licensed:

- (a) a fixed-domain theorem about localization of standing under realized persistence;
- (b) a theorem about repair locality, bounded-localization equivalence, perturbation, repair semantics, retained standing, and fail-closed continuation;
- (c) a proof that realized persistence with lawful repair semantics cannot keep standing global, diffuse, abstract, unbounded, or trace-only;
- (d) a normal-form theorem for admissibility-preserving localization regimes;
- (e) a structural bridge from realized persistence to bounded standing-localization equivalence.

1.2 Forbidden readings

The following readings are not licensed:

- (a) a proof that physical bodies, organisms, observers, or biological life are instantiated;
- (b) a proof that concrete spacetime, matter, empirical systems, or material embodiment are forced;
- (c) a claim that repair capacity is biochemical repair;
- (d) a claim that perturbation is empirical noise rather than standing-relevant disturbance;
- (e) a claim that bounded-localization equivalence selects a canonical representative;
- (f) a claim that every formal persistence relation localizes.

Standing localization is a structural consequence of realized persistence with lawful repair semantics. It proves that repairable standing-bearing persistence cannot remain unlocalized. It does not identify the localized structure with any physical or biological substrate.

2 Dependency Chain and Import Discipline

The proof uses prior AASC results as fixed structural dependencies rather than existential assumptions. The dependency chain is:

null closure $\Rightarrow$ governance induction $\Rightarrow$ continuation-locus induction $\Rightarrow$ role-anchor fixation $\Rightarrow$ persistence-regime

The following results are imported only in their fixed-domain forms.

- D1. **Null closure.** Total AASC vacuity cannot occupy the admitted side of fixed-domain admissibility-bearing evaluability.
- D2. **Governance induction.** Non-degenerate continuation induces non-vacuous constraint-bundle governance unique up to governance equivalence.
- D3. **Continuation-locus induction.** Non-degenerate standing-bearing continuation induces a continuation-locus equivalence class.
- D4. **Role-anchor fixation.** Under admissible boundary trace and internal non-selection, continuation loci induce fixed role-anchor equivalence classes.
- D5. **Realized persistence.** Irreversible standing-bearing continuation induces a realized persistence-regime equivalence class.
- D6. **Standing-admissibility identity.** On the positive side of a fixed domain, admissibility, standing, identity, transport, quotient consistency, and classificatory legitimacy collapse extensionally.
- D7. **AMetric boundary discipline.** No selector, metric, ranking, default, probability, parameter, or canonical representative may carry admissibility-bearing authority without admitted witness structure.
- D8. **Bivalence and fail-closed standing.** Evaluated fixed-domain status is admitted or not admitted; there is no third status and no same-act repair of failed standing.

- D9. **Trajectory irrecoverability.** Once standing failure enters a retained same-scope lineage, no admissible continuation restores coherence as the same evaluated act.
- D10. **Anchor–tensor–skin decomposition.** Fixed-domain admissibility-bearing content factors into anchors, tensors, and skin; skin has no gate authority or standing-bearing load.
- D11. **Fixed-domain exhaustion.** Same-scope admissibility-bearing variation factors through recognized local loci; outside them the alternative is bookkeeping, scope change, AMetric inertness, or inadmissible import.

No dependency asserts biological, empirical, material, or concrete-domain instantiation.

3 Definitions and Type Discipline

Definition 3.1 (Realized persistence regime). RealPers_D holds iff D contains a realized persistence-regime equivalence class $[\mathcal{P}_D]_{\equiv_{\text{Pers}}}$ induced by irreversible standing-bearing continuation. The class preserves role-anchor participation, trajectory behavior, transport, quotient behavior, failure behavior, consequence retention, and irreversibility profile under lawful same-domain admissibility-preserving redescription.

Definition 3.2 (Standing-relevant perturbation). Perturb_D holds iff the realized persistence regime admits a disturbance profile capable of altering standing-bearing continuation unless some admissibility-preserving boundary, repair, or localization structure preserves the relevant standing consequence. Perturbation is not empirical noise; it is standing-relevant disturbance.

Definition 3.3 (Repair capacity). RepairCap_D holds iff admissibility-bearing standing-relevant consequence degradation admits bounded lawful continuation-preserving correction without identity reset, illicit same-act repair, or unrestricted global reindexing. Repair capacity is not arbitrary update: it preserves fixed-domain identity, transport, quotient behavior, and fail-closed standing while distinguishing lawful correction from replacement, same-act repair, and envelope drift.

Definition 3.4 (Repair semantics). RepairSem_D is the role structure by which perturbation, degradation, correction, retained identity, bounded repair locality, standing preservation, and failure are distinguished on the same fixed domain. Repair semantics is required whenever standing is preserved through lawful correction rather than by inert classification, identity reset, unrestricted global reindexing, or same-act repair.

Definition 3.5 (Localization, boundedness, and localization structure). The theorem distinguishes three layers. *Localization* means that standing-relevant participation is not globally diffuse: standing consequence is evaluated at a fixed admissibility-relevant locus. *Boundedness* means that the locus admits a finite identity-retaining perturbation/repair domain for the relevant standing window, without unrestricted global reindexing. *Localization structure* means localized bounded continuation participation equipped with identity-memory, repair semantics, perturbation charge, quotient stability, and fail-closed standing behavior. Thus localization, boundedness, and localization structure are related but not identical.

Definition 3.6 (Bounded identity locus). A bounded identity locus is the minimal same-domain standing-relevant locus over which perturbation burden, identity-memory, retained repair, and failure are jointly evaluated. Boundedness means role-boundary under admissibility, not physical extension or spatial enclosure.

Definition 3.7 (Bounded standing-localization structure). A bounded standing-localization structure $\mathcal{B}_D$ is a bounded identity locus that preserves standing-bearing consequence retention under perturbation and admissibility-preserving repair. It carries boundary, identity-memory, repair semantics, quotient stability, and fail-closed standing status. It is not a substance or object.

Definition 3.8 (Bounded-localization equivalence). Two bounded standing-localization structures $\mathcal{B}_D$ and $\mathcal{B}'_D$ are equivalent, written $\mathcal{B}_D \equiv_{\text{Bound}} \mathcal{B}'_D$, iff they preserve the same boundary role, identity-memory profile, retained repair semantics, perturbation burden, standing margin, transport behavior, quotient behavior, and fail-closed status under lawful same-domain admissibility-preserving redescription. Bounded-localization equivalence is always understood modulo such redescription.

Definition 3.9 (Standing margin). For a bounded standing-localization structure B on a standing window W , write $J_B(W)$ for identity-memory retention, $R_B(W)$ for local repair throughput, and $D_B(W)$ for perturbation burden. Retained repair is

$$\tilde{R}_B(W) = J_B(W)R_B(W),$$

and the standing margin is

$$\Sigma_B(W) = \tilde{R}_B(W) - D_B(W).$$

The sign of $\Sigma_B(W)$ is the same-scope standing-relevant viability verdict after quotienting bookkeeping coordinates.

4 Standing Localization versus Concrete Embodiment

Standing localization is not embodiment. It is the localization of standing-bearing consequence retention into a bounded role locus. Realization concerns irreversible standing-bearing participation; localization concerns bounded repair-preserving persistence structure. Embodiment requires further domain-specific participation conditions not used here. Likewise, biological life is not the conclusion of this theorem. A biological reading requires a separate physical-domain interpretation identifying bounded standing-localization roles with physical structures under additional premises.

Theorem 4.1 (Bounded standing is not embodiment).

$$[\mathcal{B}_D]_{\equiv_{\text{Bound}}} \not\Rightarrow \text{PhysicalEmbodiment}_D, \quad [\mathcal{B}_D]_{\equiv_{\text{Bound}}} \not\Rightarrow \text{BiologicalLife}_D,$$

and therefore

$$[\mathcal{B}_D]_{\equiv_{\text{Bound}}} \not\Rightarrow \exists x \text{ Embodied}(x), \quad [\mathcal{B}_D]_{\equiv_{\text{Bound}}} \not\Rightarrow \exists x \text{ Physical}(x), \quad [\mathcal{B}_D]_{\equiv_{\text{Bound}}} \not\Rightarrow \exists x \text{ Life}(x).$$

Proof. A bounded-localization class preserves role-boundary, identity-memory, retained repair, perturbation burden, standing margin, transport, quotient behavior, and fail-closed status. None of those roles is typed as empirical embodiment, physical materiality, biochemical organization, organismhood, concrete-domain occupancy, or biological organization. The theorem therefore remains structural: bounded standing localization is not physical embodiment and is not biological life. $\square$

5 Realized Persistence and Standing Consequence Retention

Lemma 5.1 (Realized persistence entails consequence retention).

$$\text{RealPers}_D \rightarrow \text{ConseqRet}_D.$$

Proof. A realized persistence regime is not mere trajectory classification. It is irreversible standing-bearing continuation with admissibility-relevant consequences preserved under lawful same-domain continuation. If no standing-relevant consequence is retained, the regime is trace-only or bookkeeping. Therefore realized persistence entails consequence retention. $\square$

Corollary 5.2 (Trace-only insufficiency at the localization layer).

$$\text{TraceOnly}_D \rightarrow \neg \text{RealPers}_D.$$

Proof. Trace-only succession records order without standing-retaining lineage. Realized persistence requires irreversible standing-bearing consequence retention. Hence trace-only succession is insufficient. $\square$

6 Perturbation and Repair Semantics

Lemma 6.1 (Perturbation pressure). *If standing-relevant perturbation is in force, then retained standing with nonzero repair capacity requires repair semantics:*

$$\text{ConseqRet}_D \wedge \text{Perturb}_D \wedge \text{RepairCap}_D \rightarrow \text{RepairSem}_D.$$

Proof. Perturbation is standing-relevant precisely because it can alter standing-bearing continuation. If consequence retention persists through such disturbance and repair capacity is nonzero, the regime must distinguish the perturbation burden, the degradation, the correction, the retained identity, the bounded locality of repair, the standing verdict, and the failure case. Those distinctions are repair semantics. Without them, the correction is either inert, scope-changing, same-act repair of failure, unrestricted global reindexing, or undefined. $\square$

Remark 6.2 (Repair capacity versus forbidden same-act repair). Repair capacity is lawful continuation-level correction before standing collapse or within a retained standing-bearing lineage. It is not a same-act conversion of a standing-failed occurrence into a standing-positive occurrence. The latter remains prohibited by fail-closed standing and trajectory irrecoverability.

7 Forced Locality of Repair

Theorem 7.1 (Forced locality of repair). *If standing on a candidate locus depends on load-bearing repair external to that locus, then exactly one of the following holds:*

- (i) *standing is transferred into the locus from outside without fresh gating;*
- (ii) *the correct standing-bearing envelope is larger than the candidate locus;*
- (iii) *the external term changes no standing verdict and is bookkeeping.*

The first two are not same-domain standing localization on the candidate localization structure. The third is not load-bearing. Therefore admissible standing-preserving repair is local to the bounded-localization equivalence class.

Proof. If external repair changes standing, then standing-bearing authority not already standing on the locus performs load-bearing work for the locus. If that authority is not freshly gated, it is transfer. If the authority is part of the same standing-bearing lineage, then the envelope was incorrectly drawn and the larger envelope is the relevant locus. If the external term changes no standing verdict, it has no same-scope operator power and is bookkeeping. Fixed-domain standing conservation excludes un-gated transfer, and quotient/domain integrity excludes envelope drift presented as same-locus standing. Hence only local repair remains admissible. $\square$

Definition 7.2 (Factorizable repair). $\text{FactorizableRepair}_D$ holds iff standing-relevant perturbation and lawful correction factor through a bounded same-domain locus preserving identity-memory, perturbation burden, local repair throughput, standing margin, and fail-closed failure behavior.

Theorem 7.3 (Repair semantics factors through bounded locality).

$$\text{RepairSem}_D \rightarrow \text{RepairLocality}_D.$$

Proof. Repair semantics distinguishes degradation, correction, retained identity, bounded locality, continued standing, and failure on the same fixed domain. If these distinctions do not factor through bounded locality, the alleged repair is unrestricted global reindexing, identity reset, envelope drift, or external transfer. Each option changes the target of standing preservation or evades fail-closed repair. Therefore lawful repair semantics factors through bounded locality. $\square$

8 Repair Semantics Requires a Bounded Identity Locus

Theorem 8.1 (Bounded locus requirement).

$$\text{RepairSem}_D \rightarrow \exists_{\text{role}} \mathcal{B}_D.$$

In particular, repair semantics requires a bounded identity locus over which perturbation burden, identity-memory, retained repair, standing preservation, and failure are distinguished.

Proof. Repair semantics compares disturbed and corrected continuation states as the same standing-bearing lineage. That comparison requires a locus whose boundary fixes what is being corrected, whose identity-memory determines whether the repaired lineage is the same lineage, whose repair throughput determines the corrective contribution, whose perturbation burden is charged against the locus, and whose failure status is fail-closed. Without a bounded locus, there is no determinate target of repair. The alleged repair becomes global coordination, unrestricted reindexing, envelope import, parameter selection, or inert trace description. Hence repair semantics forces a bounded identity locus. $\square$

Theorem 8.2 (Standing-repair localization). *Every realized persistence regime with lawful repair semantics induces bounded standing localization:*

$$\text{RealPers}_D \wedge \text{RepairSem}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

Proof. Realized persistence supplies standing-bearing consequence retention. Repair semantics distinguishes degradation, correction, retained identity, bounded repair locality, standing preservation, and failure in that retained lineage. By the bounded locus requirement, those distinctions require a bounded identity locus. When the locus is evaluated under realized standing-bearing consequence retention, it is a bounded standing-localization equivalence class rather than a bare repair label. Therefore $\text{RealPers}_D \wedge \text{RepairSem}_D$ induces $[\mathcal{B}_D]_{\equiv_{\text{Bound}}}$. $\square$

Corollary 8.3 (Perturbation form).

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RepairCap}_D \rightarrow [\mathcal{B}_D]_{\equiv_{\text{Bound}}}.$$

Proof. Realized persistence entails consequence retention. By perturbation pressure, consequence retention under perturbation with nonzero repair capacity induces repair semantics. The standing-repair localization theorem then yields the bounded-localization class. $\square$

9 Standing-Margin Compression

Theorem 9.1 (Retained repair form). *Let $E(J, R)$ be an admissible retained-repair law after second-channel, transfer, amplification, unrestricted global reindexing, and same-act repair structures have been excluded. Assume:*

- (i) $E(0, R) = 0$ and $E(J, 0) = 0$;
- (ii) $E(1, R) = R$;
- (iii) for fixed J , the map $R \mapsto E(J, R)$ is locally bounded and additive;
- (iv) sequential identity-retention composes multiplicatively.

Then

$$E(J, R) = JR.$$

Proof. Local boundedness and additivity in R force $E(J, R) = \alpha(J)R$ for each fixed J . The normalization $E(1, R) = R$ gives $\alpha(1) = 1$. Multiplicative composition of identity-retention yields $\alpha(J_1 J_2) = J_1 \alpha(J_2)$. Setting $J_2 = 1$ gives $\alpha(J) = J$. Hence $E(J, R) = JR$. $\square$

Theorem 9.2 (Standing-margin uniqueness). *Any same-scope scalar or vector-valued functional used to determine whether a bounded localization structure remains standing factors, after quotienting bookkeeping coordinates and positive affine normalization, through*

$$\Sigma_B(W) = J_B(W)R_B(W) - D_B(W).$$

Proof. After standing-admissibility identity, AMetric no-second-classifier discipline, and same-scope operator closure, an admissibility-relevant viability functional may depend only on retained local repair and perturbation burden. Additivity under disjoint bounded localization structure composition, monotonicity in retained repair and burden, and boundary normalization force a positive multiple of retained repair minus burden. Vector-valued alternatives either compress to the same scalar verdict or carry bookkeeping coordinates. Therefore the standing-relevant margin is unique up to positive affine normalization. $\square$

10 No Global Standing

Definition 10.1 (Global standing claim). GlobalStand_D is a claim that standing persists under perturbation without any bounded standing-positive localization structure supporting the relevant standing-bearing consequence.

Theorem 10.2 (Global standing non-factorization).

$$\text{GlobalStand}_D \wedge \text{Perturb}_D \rightarrow \neg \text{FactorizableRepair}_D.$$

Consequently,

$$\neg \text{FactorizableRepair}_D \wedge \text{RepairCap}_D \rightarrow \text{Collapse}_D.$$

Proof. Factorizable repair requires perturbation burden, correction, identity-memory, retained standing, and failure to be assigned to a bounded lawful continuation-preserving repair domain. A global standing claim denies such a bounded standing-positive localization structure while still requiring standing retention under perturbation. The repair burden therefore cannot be factorized without nonlocal coordination, unrestricted global reindexing, envelope import, or parameter injection. These alternatives do not provide bounded lawful correction. If repair capacity is nevertheless claimed, the mismatch collapses by fixed-domain identity, fail-closed standing, and AMetric non-selection. $\square$

Theorem 10.3 (Global non-local standing exclusion).

$$\text{GlobalStand}_D \wedge \text{Perturb}_D \wedge \text{RepairCap}_D \rightarrow \text{Collapse}_D.$$

Proof. Global standing under perturbation cannot factorize repair through a bounded lawful continuation-preserving repair domain. With repair capacity in force, non-factorizability leaves only cover drift, deferred validation of non-standing parts, same-act localization repair, envelope import, or parameter injection. Cover drift violates identity-preserving stability; deferred validation violates prior gating; same-act localization repair violates fail-closed irrecoverability; envelope import violates same-domain integrity; parameter injection violates AMetric non-selection. Therefore the global standing claim collapses. $\square$

11 No Diffuse Standing

Definition 11.1 (Diffuse standing). DiffuseStand_D holds iff standing-bearing consequence is distributed across an unbounded or non-local aggregate without a bounded identity locus that carries repair semantics and standing margin.

Theorem 11.2 (Diffuse standing consequence collapse).

$$\text{DiffuseStand}_D \wedge \text{Perturb}_D \rightarrow \neg \text{StableConseqRet}_D.$$

Proof. Diffuse standing lacks a bounded identity locus to which perturbation burden, correction, identity-memory, and failure can be charged. Under perturbation, standing consequence therefore cannot be retained as the same admissibility-bearing trajectory consequence. The alleged standing either drifts with the aggregate boundary, is selected by a hidden parameter, or is imported from an envelope. In none of these cases is standing-relevant consequence stably retained on the same fixed domain. $\square$

Corollary 11.3 (Diffuse standing exclusion).

$$\text{DiffuseStand}_D \wedge \text{Perturb}_D \wedge \text{RepairCap}_D \rightarrow \text{Collapse}_D.$$

Proof. Repair capacity requires stable consequence retention through bounded lawful correction. Diffuse standing under perturbation lacks stable consequence retention. Therefore a diffuse standing claim with perturbation and repair capacity collapses into cover drift, deferred validation, localization repair, envelope import, parameter injection, or denial of the target phenomenon. $\square$

12 ATS Load: Bounded Localization Is Not Skin

Theorem 12.1 (Localized standing load).

$$[\mathcal{B}_D]_{\equiv_{\text{Bound}}} \rightarrow \text{Anchor}(\mathcal{B}_D) \vee \text{Tensor}(\mathcal{B}_D).$$

Moreover,

$$\text{SkinOnly}(\mathcal{B}_D) \rightarrow \neg \text{LocalizedStand}_D.$$

Proof. Standing-bearing perturbation repair requires anchor/tensor localization because bounded repair semantics preserves boundary, identity-memory, retained repair, perturbation burden, standing margin, transport, quotient behavior, and fail-closed status. These roles affect admissibility-bearing continuation rather than merely displaying it. By ATS exhaustion, such load is anchor or tensor content. Skin-level continuation changes presentation without preserving bounded repair semantics. Therefore skin-only presentation cannot carry localized standing. $\square$

13 AMetric Boundary and No Default Localization

Theorem 13.1 (No default localization).

$$\text{AMetric}_D \rightarrow \neg \text{DefaultLocalize}_D(\mathcal{B}_D)$$

unless additional admissibility-bearing selector structure is introduced. Any such selector is non-minimal governance load and is not part of the bounded-localization theorem.

Proof. A default localization rule selects a bounded localization structure, scale, representative, metric, or parameter as standing-bearing without deriving it from admitted standing-relevant structure. The AMetric boundary forbids selector, metric, ranking, default, or canonical representative authority without witness structure. If witness structure is supplied, the localization is no longer default; if not, no admissibility-bearing localization occurs. $\square$

14 Bounded-Localization Equivalence and Uniqueness

Theorem 14.1 (Uniqueness up to bounded-localization equivalence). *If two bounded standing-localization structures preserve the same boundary role, identity-memory profile, retained repair*

semantics, perturbation burden, standing margin, transport behavior, quotient behavior, and fail-closed status under lawful same-domain redescription, then they are equivalent under bounded-localization equivalence:

$$\mathcal{B}_D \equiv_{\text{Bound}} \mathcal{B}'_D.$$

Proof. Those listed roles exhaust the admissibility-bearing content of standing localization. If two bounded localization structures preserve all of them, no remaining same-scope standing-bearing difference is available. Any further descriptive difference is skin, bookkeeping, or cross-domain load. Therefore the structures are equivalent under $\equiv_{\text{Bound}}$. $\square$

Corollary 14.2 (No canonical representative).

$$[\mathcal{B}_D]_{\equiv_{\text{Bound}}} \not\Rightarrow \exists! B.$$

A bounded-localization equivalence class need not determine a privileged representative. Representative selection requires additional admissibility-bearing selector structure.

15 Finite Witness Model

Let M contain three standing-relevant stages s_0, s_1, s'_1 , one perturbation p , one lawful repair map r , identity-memory J , local repair throughput R , perturbation burden D , standing margin $\Sigma = JR - D$, quotient relation $\sim$, and fail-closed standing predicate Stand .

The intended continuation is:

$$s_0 \xrightarrow{p} s_1 \xrightarrow{r} s'_1.$$

The continuation is standing-preserving only if the same bounded localization structure retains identity-memory, absorbs perturbation burden, supplies local repair, and satisfies the standing margin after quotient-respecting redescription.

Theorem 15.1 (Finite localization witness). *In M ,*

$$\text{RealPers}(M) \wedge \text{Perturb}(M) \wedge \text{RepairCap}(M) \rightarrow [\mathcal{B}_M]_{\equiv_{\text{Bound}}}.$$

Proof. Consider the collapse cases. Removing perturbation removes the standing-relevant disturbance that exposes localization pressure. Removing repair capacity prevents standing from being preserved through disturbance by admissible correction. Removing bounded repair locality turns correction into unrestricted global reindexing and destroys standing-bearing identity preservation. Removing the bounded locus leaves no determinate target of repair, burden, or identity-memory. Removing the identity boundary prevents repair from distinguishing same-locus retention from replacement or envelope drift. Removing quotient discipline converts harmless presentation drift into hidden standing load. Removing fail-closed standing permits failed standing to be silently repaired as the same evaluated act. Removing irreversibility converts consequence retention into reindexable trace succession. Removing ATS load leaves the alleged localization structure skin-only and without standing-bearing authority.

Each removal destroys standing-bearing localization. Restoring standing-bearing continuation under perturbation and repair capacity restores precisely the bounded localization structure roles. Hence the finite model induces $[\mathcal{B}_M]_{\equiv_{\text{Bound}}}$. $\square$

Corollary 15.2 (Witness synthesis). *The finite witness shows that bounded standing localization is not reducible to trace succession, global repair, diffuse aggregation, unrestricted global reindexing, or presentation persistence. It requires bounded identity, local repair, perturbation charge, quotient stability, fail-closed standing, and retained consequence.*

16 Bounded-Localization Normal Form

Theorem 16.1 (Bounded-localization normal form). *Every admissibility-preserving localization regime factorizes through:*

- (i) realized persistence-regime equivalence $[\mathcal{P}_D]_{\equiv \text{Pers}}$;
- (ii) standing-bearing consequence retention;
- (iii) standing-relevant perturbation exposure;
- (iv) bounded repair locality and repair semantics RepairSem_D ;
- (v) bounded identity locus with boundary, identity-memory, local repair, and fail-closed failure;
- (vi) bounded standing-localization equivalence $[\mathcal{B}_D]_{\equiv \text{Bound}}$.

Symbolically,

$$\text{RealPers}_D \wedge \text{RepairSem}_D \Rightarrow_{\text{struct}} ([\mathcal{P}_D]_{\equiv \text{Pers}}, \text{RepairLocality}_D, [\mathcal{B}_D]_{\equiv \text{Bound}}),$$

and the perturbation form follows when $\text{Perturb}_D \wedge \text{RepairCap}_D$ induces RepairSem_D .

Proof. Realized persistence supplies consequence retention. Standing-relevant perturbation exposes whether consequence retention can survive disturbance. Lawful repair capacity requires repair semantics, and repair semantics requires bounded repair locality rather than global reindexing. The bounded identity locus fixes what is degraded, corrected, retained, and failed. Standing margin compression fixes the admissibility-relevant repair/burden profile. Forced locality excludes external repair import. No-global and no-diffuse standing exclude unlocalized alternatives. ATS excludes skin-only localization. AMetric discipline excludes default localization selection. What remains is bounded-localization equivalence. These components form the normal form. $\square$

17 Main Theorem: Standing-Repair Localization

Theorem 17.1 (Standing-repair localization theorem). *Let D be a fixed AASC domain. If D supports a realized persistence regime with lawful repair semantics, then standing cannot remain global, abstract, diffuse, trace-level, or unlocalized. It induces a bounded standing-localization equivalence class:*

$$\text{RealPers}_D \wedge \text{RepairSem}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

Proof. Assume $\text{RealPers}_D \wedge \text{RepairSem}_D$. Realized persistence supplies standing-bearing consequence retention. Repair semantics distinguishes degradation, correction, retained identity, bounded repair locality, standing preservation, and failure in that lineage. By the bounded locus requirement, repair semantics requires a bounded identity locus. By forced locality, load-bearing repair cannot be

supplied externally without transfer, envelope drift, or bookkeeping. By standing-margin compression, same-scope standing verdicts factor through retained local repair minus perturbation burden where perturbation exposure is present. By the global standing non-factorization theorem and diffuse standing consequence collapse, unlocalized alternatives cannot sustain lawful repair. By ATS, the bounded locus must be anchor/tensor content rather than skin. By AMetric no-default localization, it is not selected by arbitrary representative or parameter. Therefore the standing-bearing localization induced is a bounded standing-localization equivalence class $[\mathcal{B}_D]_{\equiv \text{Bound}}$. $\square$

Corollary 17.2 (Perturbation-and-repair form).

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RepairCap}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

Proof. By realized persistence, consequence retention is in force. By perturbation pressure, consequence retention under perturbation with nonzero repair capacity induces repair semantics. The standing-repair localization theorem then yields $[\mathcal{B}_D]_{\equiv \text{Bound}}$. $\square$

18 No Life or Ontology Overclaim

Proposition 18.1 (Life boundary).

$$[\mathcal{B}_D]_{\equiv \text{Bound}} \not\Rightarrow \exists x \text{ Life}(x).$$

The theorem proves bounded standing localization. It does not identify the bounded localization structure with biological life, organismic structure, biochemical repair, metabolism, heredity, intelligence, or empirical bodyhood.

Proof. The bounded localization structure has structural roles: boundary, identity-memory, retained repair, perturbation burden, standing margin, quotient stability, and fail-closed standing. Biological life requires a physical-domain interpretation of those roles plus substrate-specific conditions. Those conditions are not part of the theorem. $\square$

19 Route Exhaustion

A same-domain attempt to preserve standing under realized persistence and repair semantics without bounded standing localization must instantiate at least one of the following failure families.

Route	Collapse mechanism
Global standing	no bounded standing-positive localization structure; repair cannot factorize.
Diffuse standing	stable consequence retention fails under perturbation.
External repair	transfer, envelope drift, or bookkeeping.
Bounded repair locality failure	unrestricted global reindexing destroys standing-bearing identity preservation.
Deferred validation	unstanding parts enter standing-bearing continuation before gating.
Same-act localization repair	violates fail-closed irrecoverability.
Parameter injection	violates AMetric non-selection.
Skin-only localization structure	lacks anchor/tensor standing-bearing load.
Representative selection	adds non-minimal selector structure.
Trace-only persistence	lacks standing-bearing consequence retention.

No further same-domain route remains: any further alternative either changes no standing verdict, changes domain, imports selector load, or denies the target phenomenon.

20 Objection Matrix

Objection	Reply
The result proves biological life.	No. It proves bounded standing-localization equivalence, not biochemical or organismic life.
The result proves physical bodies.	No. Boundedness is role-structural; bounded standing is not physical embodiment.
Repair smuggles biology.	Repair means bounded lawful continuation-preserving correction under perturbation, without identity reset, same-act repair, or global reindexing.
Perturbation is empirical noise.	Perturbation means standing-relevant disturbance.
Global standing can persist.	Global standing cannot factorize repair through a bounded lawful repair domain.
Diffuse standing can persist.	Diffuse standing lacks stable consequence retention under perturbation.
Repair violates no same-act repair.	Lawful repair capacity preserves retained lineage; same-act repair of standing failure remains forbidden.
Localization selects a representative.	The theorem induces a bounded-localization equivalence class, not a canonical representative.
The localization structure is skin-only.	ATS assigns standing-bearing localization to anchor/tensor content.
The theorem assumes concrete reality.	It proves structural localization within realized persistence; concrete-domain participation is not assumed.

21 Closure Ledger

Escape locus	Closure result
Trace-only succession	no standing-bearing consequence retention.
Global standing	no factorizable repair under perturbation.
Diffuse aggregate	no stable consequence retention under perturbation.
External repair	transfer, scope drift, or bookkeeping.
Global reindexing	destroys standing-bearing identity preservation.
Same-act repair	excluded by fail-closed irrecoverability.
Default localization	excluded by AMetric no-selector discipline.
Skin-only boundary	no standing-bearing authority.
Representative privilege	additional selector load, not minimal localization.
Biological reading	outside theorem type.
Concrete reading	outside theorem type.

22 Canonical Results

22.1 Canonical main theorem

$$\text{RealPers}_D \wedge \text{RepairSem}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

22.2 Canonical perturbation-and-repair form

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RepairCap}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

22.3 Canonical normal form

Every admissibility-preserving localization regime factorizes through realized persistence-regime equivalence, consequence retention, perturbation exposure, bounded repair locality, repair semantics, bounded identity locus, and bounded standing-localization equivalence.

22.4 Canonical anti-overclaim

The theorem proves bounded standing localization under realized persistence and lawful repair semantics. It does not prove concrete reality, physical embodiment, biological life, organisms, observers, spacetime, or material systems.

22.5 Canonical repair formulation

Repair capacity is bounded lawful admissibility-preserving correction under perturbation. It is not identity reset, unrestricted global reindexing, or same-act repair of a standing-failed evaluated act.

22.6 Canonical standing-margin formulation

Up to positive affine normalization and bookkeeping coordinates, the same-scope standing margin is

$$\Sigma_B(W) = J_B(W)R_B(W) - D_B(W).$$

22.7 Canonical no-global-standing formulation

Standing-bearing repair under perturbation cannot be global or diffuse without transfer, envelope drift, parameter injection, same-act repair, deferred validation, non-factorizable repair, or collapse.

23 Short Proof Form

1. Realized persistence entails standing-bearing consequence retention.
2. Repair semantics distinguishes standing-relevant degradation, perturbation burden, correction, retained identity, bounded repair locality, standing preservation, and failure.
3. Perturbation plus nonzero repair capacity is one route by which repair semantics is induced.
4. Repair semantics requires bounded repair locality and a bounded identity locus.
5. Load-bearing repair external to the locus is transfer, envelope drift, global reindexing, or bookkeeping.
6. Same-scope standing viability compresses to retained local repair minus perturbation burden.
7. Global standing fails by non-factorizable repair.
8. Diffuse standing fails by loss of stable consequence retention.
9. ATS excludes skin-only bounded localization structures.
10. AMetric discipline excludes default localization and canonical representative selection.
11. Therefore realized persistence with lawful repair semantics induces a bounded standing-localization equivalence class.

24 Conclusion

Repair locality is the organizing constraint of standing localization. Realized standing-bearing persistence cannot remain merely trace-level, global, diffuse, abstract, or unlocalized when lawful repair semantics are in force. Perturbation exposes standing-retention load. Repair capacity is bounded lawful correction rather than arbitrary update, identity reset, same-act repair, or unrestricted global reindexing, and it requires repair semantics. Repair semantics requires bounded repair locality and a bounded identity locus. Fixed-domain standing conservation blocks external repair transfer and envelope drift. AMetric discipline blocks default localization and representative selection. ATS blocks skin-only localization authority. The standing-relevant viability verdict compresses to retained local repair minus perturbation burden.

The resulting theorem is:

$$\text{RealPers}_D \wedge \text{RepairSem}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

The perturbation-and-repair form is:

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RepairCap}_D \rightarrow [\mathcal{B}_D]_{\equiv \text{Bound}}.$$

Thus repair locality is the structural source of bounded standing localization: every admissibility-preserving localization regime of this type factorizes through realized persistence-regime equivalence, consequence retention, perturbation exposure, bounded repair locality, repair semantics, bounded identity locus, and bounded standing-localization equivalence. The conclusion is structural. It is not a proof of biological life, physical bodies, empirical spacetime, organisms, observers, or concrete ontology. It establishes the bounded-localization layer required by realized persistence with lawful repair semantics.

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